BAJAJ ALLOYS, BHIWANI

BUSHES

Complete Industry & Product Report: Product, Types, Manufacturing, Market, B2B / B2C

Prepared by: Intern, Business Research
Date: October 2026
Status: Draft for management review

Contents

- 1. Executive summary
- 2. What is a bush?
- 3. Types of bushes
- 4. Materials
- 5. Manufacturing process
- 6. Quality and standards
- 7. Market size & growth
- 8. B2B and B2C market
- 9. Competition
- 10. Opportunity & recommendations for Bajaj Alloys
- 11. Data notes & sources

1. Executive summary

A bush (also called bushing or sleeve bearing) is a cylindrical sleeve placed between a rotating or sliding shaft and its housing. It reduces friction and wear, absorbs vibration and is cheaper to replace than the shaft or housing. Bronze, gunmetal, phosphor bronze and brass bushes are classic products of a copper-alloy foundry.

- **Market proxy:** No source gives a clean India figure for bronze bushes. Proxies: India automotive bearings about USD 503 million in 2024 growing 4.15% a year to about USD 726 million by 2033 (IMARC). Global automotive bushing market USD 172 billion (2025) to USD 218 billion (2030), CAGR 4.83% (Mordor Intelligence).
- **Demand:** Automobiles, tractors, pumps, motors, compressors, steel plants, earth-moving machinery, railways, textile and sugar machinery.
- **Strength of aftermarket:** Bushes wear out and are replaced, so replacement demand is steady. One report puts India bearing aftermarket growth near 14.7% a year.
- **Recommendation:** Focus on centrifugally cast and machined bronze/gunmetal bushes for pumps, motors, tractors and heavy machines; sell to OEMs and through industrial distributors; add standard-size catalogue for retail.

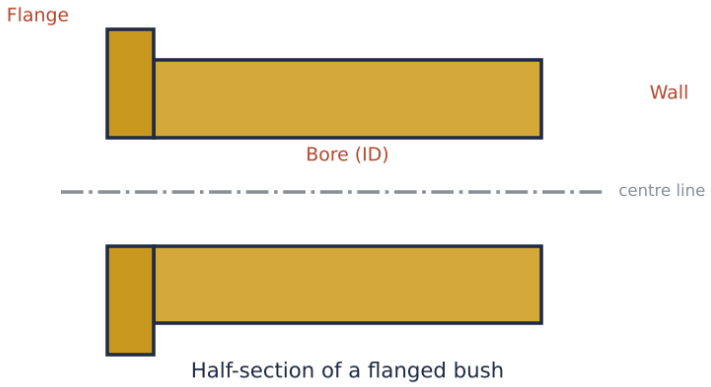


Figure 1: Half-section of a flanged bush

2. What is a bush?

A bush is a hollow cylinder, plain or flanged, press-fitted in a housing so the shaft turns inside it. The softer bush metal wears instead of the costly shaft, and a film of oil or grease keeps surfaces apart.

Functions

- Reduce friction between moving parts.
- Carry radial (and with flange, axial) loads.
- Dampen noise and vibration (rubber and polymer bushes).
- Act as replaceable wear parts.
- Guide and align shafts, pins and pivots.

Table 1: Basic terminology

Term	Meaning

Bush / bushing	Sleeve-type bearing, no rolling elements
ID / OD / length	Inner diameter, outer diameter, length
Interference fit	OD slightly bigger than housing so it is pressed in
Running clearance	Gap between shaft and bush bore, filled with lubricant
PV value	Pressure x velocity limit of the bush material

Bush vs ball bearing

Table 2: Comparison

Feature	Bush (plain bearing)	Ball/roller bearing
Friction	Higher; needs lubrication	Low
Load	High shock and heavy load	Moderate, precision
Cost	Low	Higher
Space	Compact	Larger
Noise	Quiet	Can be noisy
Life/replacement	Easy and cheap	Costlier

3. Types of bushes

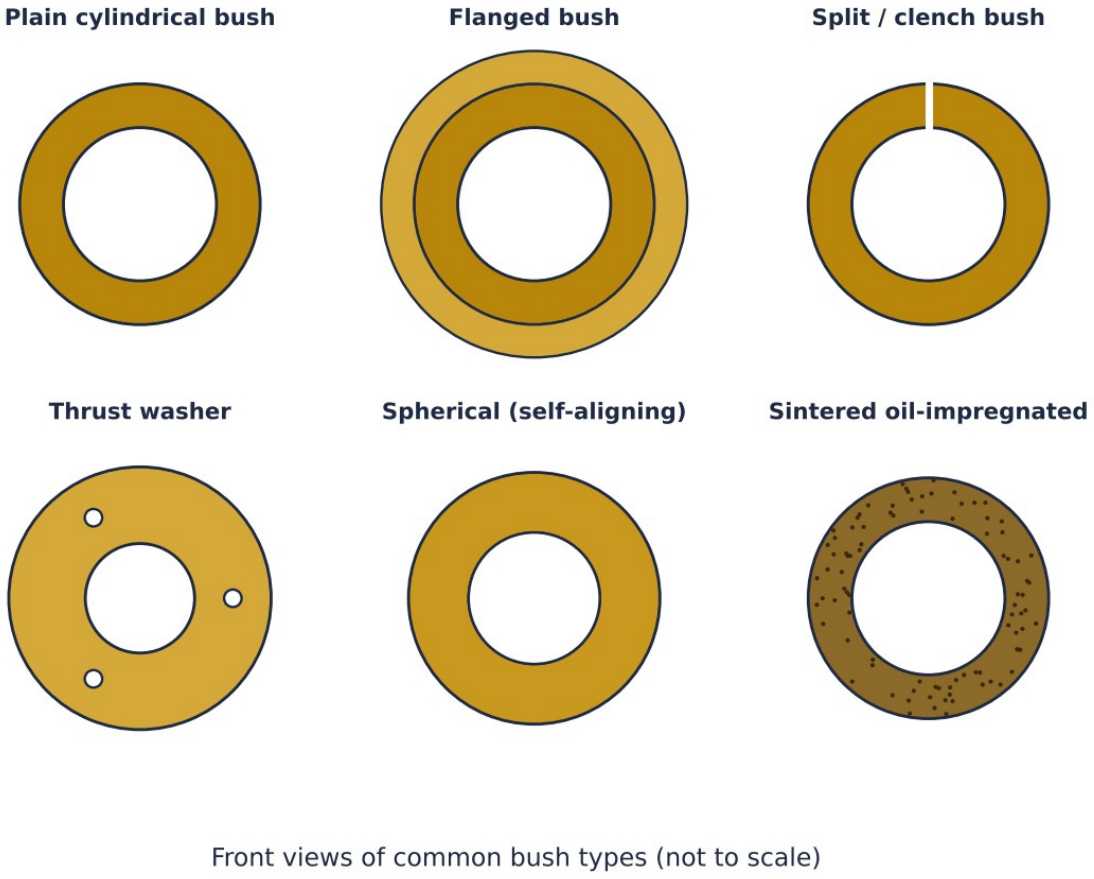


Figure 2: Front views of six common bush types

Table 3: Bush types

Type	Description	Used in

Plain cylindrical	Simple sleeve	Pumps, gearboxes, hinges
Flanged	Collar takes axial load	Automotive, machine tools
Split / clench	Rolled strip bush with seam	Engines, compact assemblies
Thrust washer	Flat ring for axial load	Gearboxes, transmissions
Spherical / self-aligning	Ball-shaped seat allows misalignment	Linkages, suspension
Sintered (oil-impregnated)	Porous metal holds oil; maintenance free	Fans, appliances, small motors
Bi-metal / composite	Steel back with bronze or polymer lining	Heavy machinery, hydraulics
Rubber / polyurethane	Elastomer bush for vibration	Suspension, stabiliser bars
Graphite plugged bronze	Solid lubricant plugs in bronze	High temp, slow heavy loads, steel plants

By material

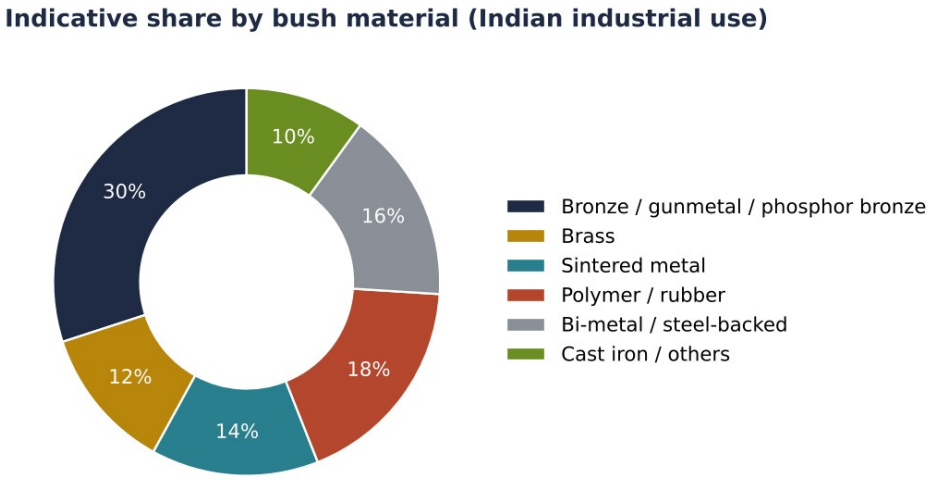


Figure 3: Indicative material split (intern estimate)

By application

- **Automotive:** suspension, engine, steering, gearbox
- **Agriculture:** tractors, harvesters, pumps
- **Industrial:** pumps, compressors, motors, gearboxes, presses
- **Heavy equipment:** excavators, cranes, mining
- **Railways:** wagons, locomotives, axle boxes
- **Electrical:** fans, appliances, small motors

4. Materials

Table 4: Copper alloys for bushes

Alloy	Typical composition	Properties	Typical use
Leaded gunmetal (LG2 / LTB2)	Cu 85, Sn 5, Pb 5, Zn 5	Good wear, pressure-tight	Pump and valve bushes
Phosphor bronze (PB1 / PB2)	Cu 88 to 90, Sn 10 to 12, P 0.2	Hard, fatigue-resistant	Gears, worm wheels, heavy loads
Leaded tin bronze (LB 5)	Cu 80, Sn 10, Pb 10	Excellent for high speed, low friction	Engine and motor bearings

Aluminium bronze	Cu 80 to 88, Al 8 to 11, Fe, Ni	Very strong, corrosion resistant	Heavy load, marine
Brass (HB/SB)	Cu 60, Zn 40	Economical, medium load	Light duty bushes
SAE 660 bearing bronze	Cu 83, Sn 7, Pb 7, Zn 3	Standard continuous-cast bronze	General industrial

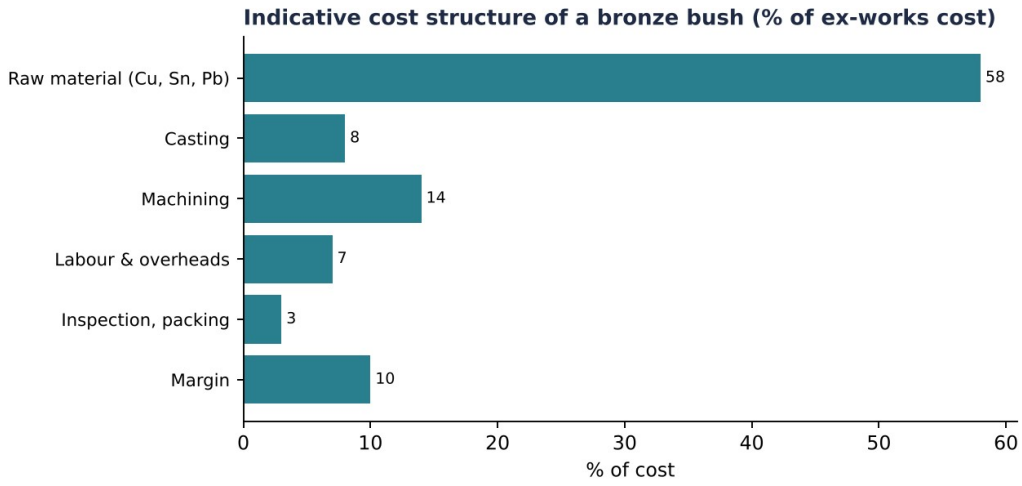


Figure 4: Indicative cost structure (illustrative)

Tin is expensive and volatile, so alloy choice (for example LG2 vs PB2) has a large effect on cost. Copper and bronze scrap recycling at Bajaj Alloys gives a cost advantage.

5. Manufacturing process

Bronze bushes are mainly made by (a) centrifugal casting of hollow tubes, (b) continuous casting of rods and tubes, (c) sand/permanent mould casting for special sizes, (d) powder metallurgy for sintered bushes, and (e) rolling from strip for split bushes.

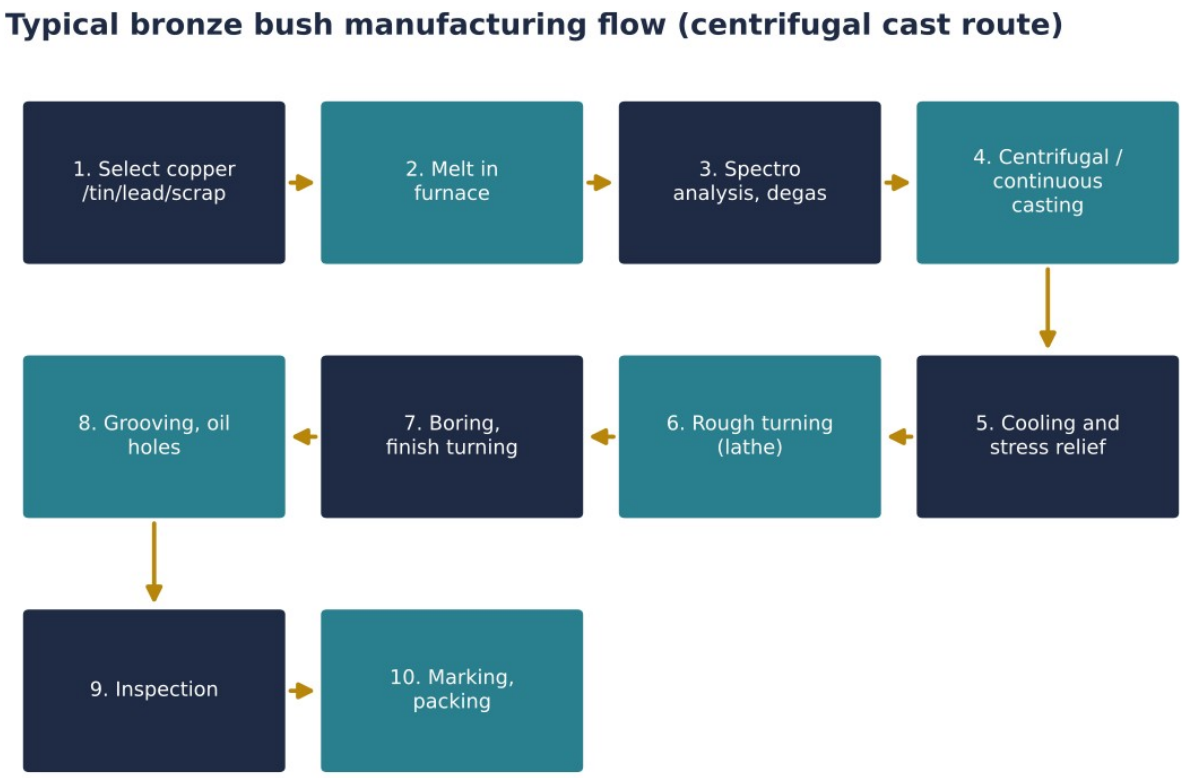


Figure 5: Process flow for a cast bronze bush

- **Step 1: Charge preparation:** Virgin copper, tin, lead, zinc and clean scrap are weighed to the target alloy.
- **Step 2: Melting and refining:** Induction/crucible furnace at 1,100 to 1,200 degrees C, flux and degassing, then sample to the spectrometer to adjust the composition.
- **Step 3: Casting:** Centrifugal casting: metal is poured into a spinning mould (about 600 to 1,500 rpm); centrifugal force gives dense, defect-free walls. Continuous casting draws hollow rods through a graphite die.
- **Step 4: Machining:** Tubes are cut, turned on CNC lathes to OD, bored to ID, faced to length, chamfered. Flanges, oil grooves and holes are added.
- **Step 5: Sintering route (alternative):** Metal powder is compacted in a die, sintered at about 800 degrees C, sized, and vacuum-impregnated with oil.
- **Step 6: Inspection:** Dimensional (go/no-go gauges), hardness, visual check for porosity, material certificates.
- **Step 7: Packing:** Oiled, VCI paper, boxed by size and lot.

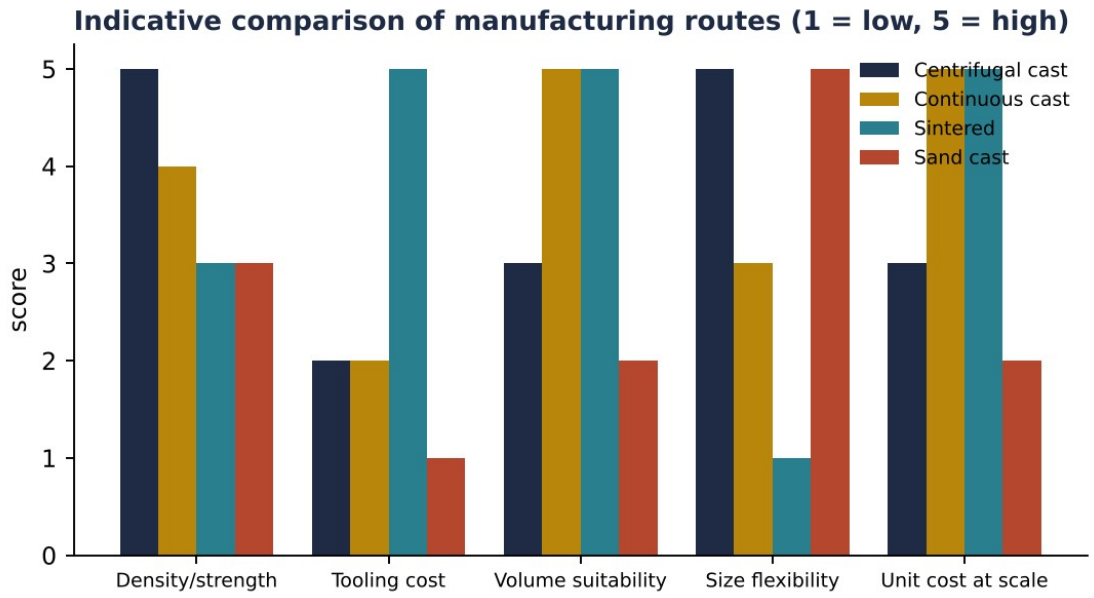


Figure 6: Qualitative scoring of routes (intern judgement)

6. Quality and standards

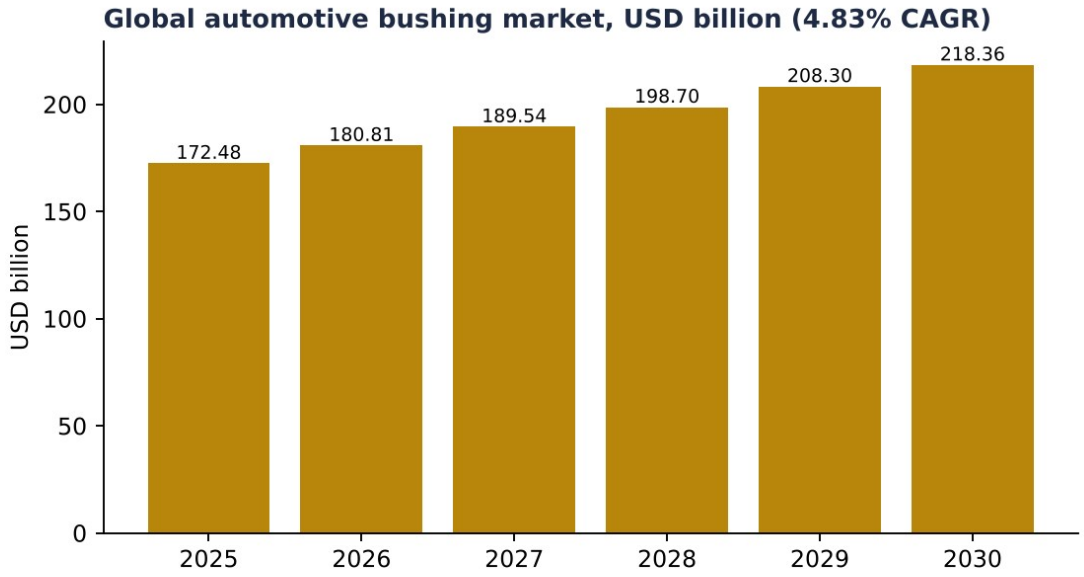
Table 5: Standards for bushes

Standard	Scope
IS 318	Leaded tin bronze ingots and castings
IS 28	Phosphor bronze ingots and castings
IS 7598 / IS 11620	Plain bearing related standards
IS 4378	Bearing bush related guides
ASTM B505 / SAE 660	Continuous cast bronze
DIN 1494 / ISO 3547	Dimensions of plain bearing bushes
ISO 9001 / IATF 16949	Quality systems, automotive customers need IATF

Typical customer demands: chemical composition certificate, hardness (HB 60 to 120), dimensional tolerance (H7 bore, interference-fit OD), surface roughness, PPAP for automotive customers.

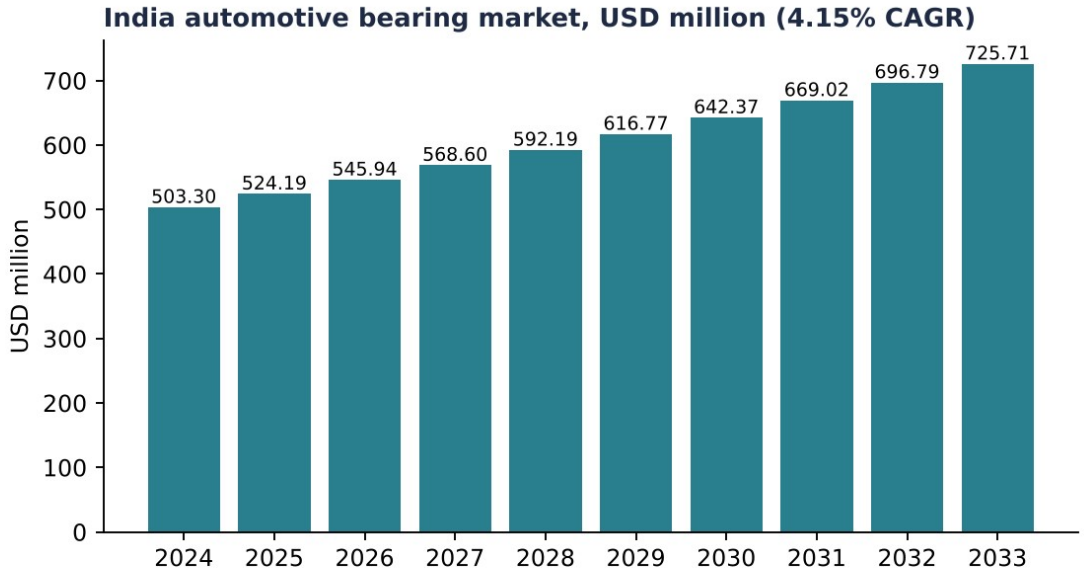
7. Market size and growth

7.1 Global and Indian proxies



Source: Mordor Intelligence (2025: USD 172.48 bn, 2030: USD 218.35 bn); interim years interpolated at stated CAGR.

Figure 7: Global automotive bushing market



Source: IMARC Group (2024: USD 503.3 m, 2033: USD 725.5 m); interim years interpolated at stated CAGR.

Figure 8: India automotive bearing market, a proxy for bush demand

Table 6: Published figures used as proxies

Indicator	Value	Source
Global automotive bushing 2025	USD 172.48 bn	Mordor Intelligence
Global 2030	USD 218.35 bn (CAGR 4.83%)	Mordor Intelligence
India automotive bearing 2024	USD 503.3 m	IMARC

India automotive bearing 2033	USD 725.5 m (CAGR 4.15%)	IMARC
India bearings overall growth	3.2% (Technavio 2024 to 2029) to 9.7% (IMARC 2026 to 2034)	Technavio, IMARC
India bearing aftermarket growth	About 14.7% a year	Astute Analytica

Important: the global figure covers all automotive bushings (including rubber and polymer, and complete assemblies), so it is far bigger than the metal-bush segment a foundry serves. Treat it as direction of growth, not as Bajaj Alloys' addressable market. A bottom-up estimate (vehicles produced x bushes per vehicle x average price) should be built next.

7.2 Bottom-up illustration

Table 7: Method for building an addressable market (inputs to be validated)

Item	Illustrative value
India vehicle production (all categories, approx.)	About 28 to 30 million a year
Metal / bronze bushes per vehicle (average, varies hugely)	5 to 20 pieces
Average price per bush	INR 20 to 150
Tractors, pumps, motors, industry	Additional millions of pieces a year

7.3 Demand drivers

- Rising vehicle and tractor production; EV shift reduces engine bushes but increases suspension and e-axle bushes.
- Pumps, motors and capex in water, steel, cement, power.
- Maintenance and replacement (aftermarket) growth.
- Import substitution and China+1 for machine components.
- Railways and defence indigenisation.

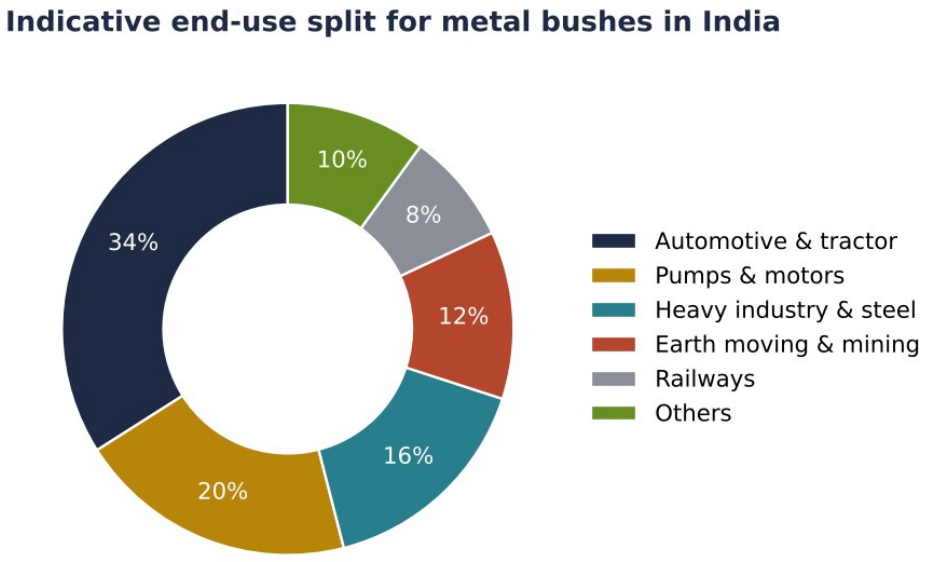


Figure 9: Indicative end-use split (intern estimate)

7.4 Regional clusters

- Ludhiana, Jalandhar (Punjab): auto components
- Rajkot, Ahmedabad (Gujarat): engineering castings, bearings
- Pune, Chennai, Coimbatore: OEM and pumps
- Gurugram, Faridabad, Bhiwani: NCR auto cluster, brass and bronze foundries

8. B2B and B2C market

8.1 B2B

Most bush sales are B2B: OEM supply to vehicle, tractor, pump and motor manufacturers; contracts with steel plants, railways and heavy machinery companies; and sales to industrial distributors.

8.2 B2C and retail aftermarket

B2C means mechanics, garages and individual owners buying replacement bushes (suspension, hinge, tractor and pump bushes) from spare-part shops or online. It is small in value for a foundry but offers higher margins and brand-building.

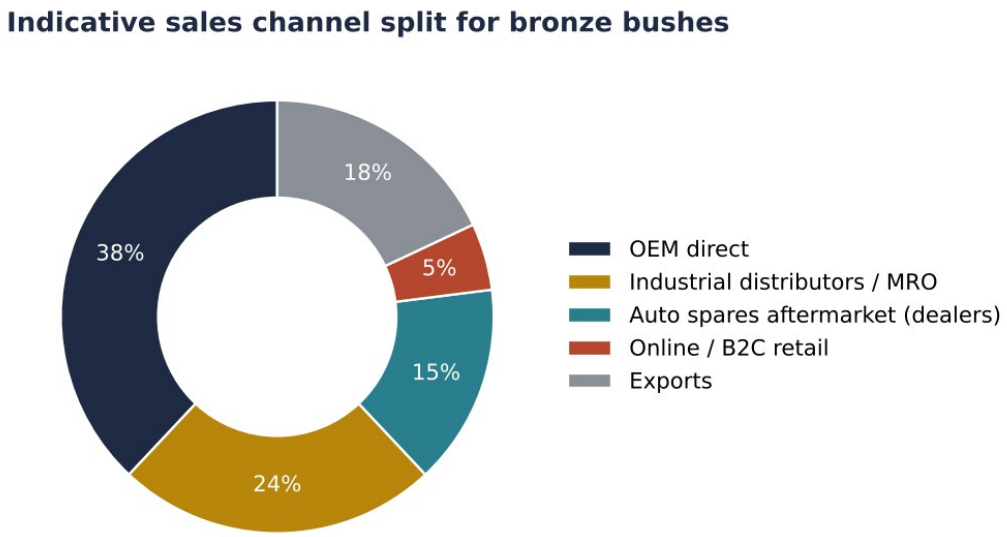


Figure 10: Indicative channel mix (illustrative)

Table 8: B2B vs B2C comparison for bushes

Factor	B2B (OEM / industrial)	B2C (aftermarket / retail)
Buyers	Auto OEMs, tractor makers, pump and motor makers, steel plants, railways	Mechanics, garages, farmers, small workshops
Order size	Thousands per month, annual contracts	Single pieces or small packs
Requirements	Drawing, PPAP, IATF, test certificates	Fit, availability, price, brand
Pricing	Cost-plus with metal price pass-through (LME copper)	MRP-based, margin for retailer
Margin	10 to 18%	25 to 40% (with distributor margins)
Payment	45 to 90 days	Cash / short credit
Marketing	Key-account visits, approvals, trials (6 to 12 months)	Catalogue, mechanic meets, e-commerce, IndiaMART

Risk	Customer concentration, price squeeze	Spurious competition, many SKUs

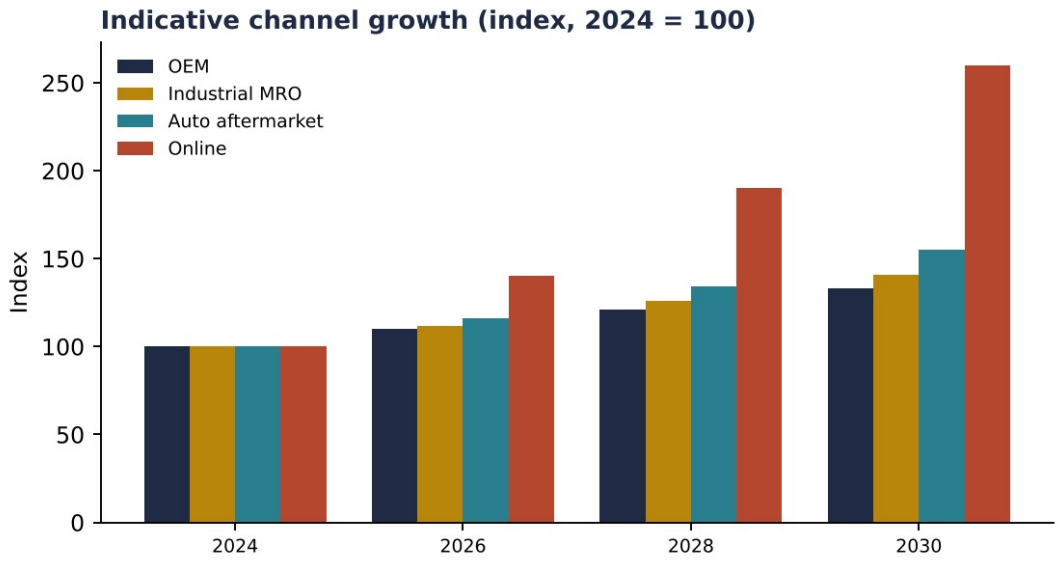


Figure 11: Illustrative channel growth (aftermarket and online faster than OEM)

9. Competition

Table 9: Competitive landscape (names for orientation only)

Segment	Players (examples)
Organised bearings and bushes	Schaeffler India, SKF India, NBC (National Engineering Industries), Timken India, FAG; Federal-Mogul/Tenneco bearings
Metal bush / bronze specialists	Many small and mid-sized foundries in Punjab, Gujarat, Maharashtra, Haryana
Sintered	Sundaram-Clayton group units, Hindustan Powder Metallurgy-type units and imports
Global	GGB, Saint-Gobain, Kolbenschmidt, Oilite, Igus (polymer)

The metal bush segment is fragmented and unorganised. Differentiation: consistent alloy chemistry, tight dimensions, certificates, and on-time supply.

10. Opportunity and recommendations for Bajaj Alloys

- Use in-house scrap and melting to produce continuous-cast and centrifugally cast bronze/gunmetal tube and rod: sell both as raw bush stock and as finished machined bushes.
- Target pump, motor and tractor OEMs in Haryana, Punjab, Rajasthan first.
- Add a standard catalogue (e.g. 50 common sizes) for distributors and e-commerce.
- Invest in CNC lathes, bore gauges and a spectrometer; pursue ISO 9001 and later IATF 16949.
- Hedge copper and tin price exposure through pricing formulas linked to LME/MCX.
- Risks: tin price swings, EV shift, Chinese imports, customer concentration.

Table 10: Indicative roadmap

Item	Action	Timeline
Product range and costing	10 alloys/sizes, cost sheets	Month 1
Casting set-up	Centrifugal / continuous caster trial	Months 2 to 5
Machining cell	2 CNC lathes, gauges	Months 3 to 6
Customer trials	5 OEM and 10 distributors	Months 6 to 10
Certification	ISO 9001 / IATF	Months 6 to 12

11. Data notes and sources

Market numbers come from published research summaries (Mordor Intelligence for global automotive bushings, IMARC Group for India automotive bearings and India bearings, Technavio and Astute Analytica for India bearings growth). No source gave a clean India figure for bronze bushes specifically, so these are used as proxies, and the report says so. Interim yearly values in charts are interpolated at the stated CAGR. Charts labelled 'indicative' are intern estimates built from typical industry structure; they are not audited statistics and must be validated with plant and customer data. Diagrams are schematic drawings made for this report, not photographs.

- Mordor Intelligence: Automotive Bushing Market
- IMARC Group: India Automotive Bearing Market; India Bearings Market
- Technavio: India Bearings Market Analysis
- Astute Analytica: India Bearings Market
- BIS standards IS 318, IS 28; ISO 3547